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Dendritic growth dynamics and morphology control in the Amplitude-expanded Phase Field Crystal Model with Gaussian colored noise

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ABSTRACT

The Amplitude-expanded Phase Field Crystal (APFC) model is employed to systematically investigate the dynamics of dendritic growth under different undercooling conditions and to explore in depth the role of Gaussian colored noise — a ubiquitous phenomenon in both natural and engineering environments — in regulating crystal growth morphology. The results show that the competition between interface energy anisotropy and interface kinetic anisotropy is the key driving force behind the formation of different dendritic morphologies. Gaussian colored noise can selectively amplify perturbations that match the system's unstable spectrum and control the crystal growth direction, significantly enhancing interfacial instability and enabling control over the complexity of dendritic morphology. For the first time within the APFC framework, a quantitative mapping relationship among noise parameters (intensity, filter width, characteristic wavenumber) interfacial stability, and dendritic morphology has been established. This offers a new theoretical perspective on the interfacial evolution mechanism in non-equilibrium solidification processes and provides a new dimension to control morphology during crystal growth.

1. Introduction

Dendritic growth is one of the most representative and universal phenomena in materials science, profoundly influencing material properties and performance in manufacturing processes such as casting, welding, and additive manufacturing. The morphology and evolution of dendritic growth are primarily influenced by various key factors, including undercooling, interface energy anisotropy, solute concentration, and heat transfer. The coupling of these factors determines the stability of the solid–liquid interface and the evolution pathways of dendrites, making precise modeling and simulation essential for understanding crystal growth mechanisms and interface dynamics [1–3]. Studies have shown that under different undercooling conditions, dendritic growth undergoes a typical evolution process, transitioning from faceted structures to dendritic patterns, and eventually to spherical morphologies. Understanding these complex growth modes has long been a central challenge in solidification research. In principle, accurate theoretical prediction of such morphological evolution requires the simultaneous consideration of atomic-scale behavior and mesoscopic interfacial dynamics.

Over the past two decades, phase-field (PF) models have been widely used to simulate complex interfacial evolution processes, such as dendritic and alloy solidification. These models have successfully revealed a variety of morphologies observed in experiments [4–6]. PF models employ a continuous order parameter to describe the solid–liquid interface, which allows for the natural treatment of complex interface morphologies, nucleation, and growth. However, due to the lack of explicit lattice structure information, capturing microscopic mechanisms such as atomic arrangements, lattice symmetry, defect evolution, and elastic effects is difficult. This limitation restricts the application of PF models to exploring the mesoscopic physical essence of solidification.

To address this limitation, Elder et al. [7] proposed the phase-field crystal (PFC) model. By representing atomic number density as a periodic field, the PFC model enables studies of crystal structures, defect evolution, and solid–liquid interface dynamics on diffusive time scales. In 2014, Tang et al. [8] employed PFC modeling to investigate morphology transitions in crystal growth. They demonstrated how the competition between interface energy anisotropy and interface kinetics anisotropy shapes dendritic growth patterns and constructed a morphological phase diagram of crystal growth, providing important insights into crystal growth mechanisms under non-equilibrium regime.

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In addition, Chen et al. [9] used a single-mode PFC model to explore morphological transitions in faceted and dendritic crystal growth, and examined the influence of noise on the crystalline structures in a wide range of control parameters. However, while the PFC model can reach diffusive time scales, its need to resolve atomic-scale density fluctuations imposes a resolution constraint smaller than the lattice spacing, which drastically increases computation cost as the system size increases, thereby limiting simulations of large-scale solidification processes.

Building on this, Goldenfeld et al. [10,11] proposed the Amplitude-expanded Phase Field Crystal (APFC) model in 2005. The APFC model captures only the slowly varying amplitude field by expanding the density in the PFC model with complex amplitudes, thus avoiding the need to directly resolve the rapidly oscillating density field. This approach relaxes the requirement for atomic-scale resolution, thereby greatly improving computational efficiency. Through continuous development and refinement [12–16], the APFC model has demonstrated distinct advantages in addressing problems such as crystal solidification, dislocation evolution, and interface migration. Compared with the traditional PF model, APFC bridges the atomic and mesoscopic scales while retaining lattice details, making it more suitable for studying microscopic mechanisms. In contrast to the PFC model, APFC circumvents the resolution constraints imposed by atomic-scale density fluctuations, thereby enabling simulations of diverse dendritic growth dynamics in large-scale systems [12,17]. However, the ability of the APFC model to describe the morphological evolution of dendrites during solidification has not yet been systematically validated, which directly limiting its applicability in materials design.

Moreover, during practical solidification and crystal growth processes, the solid–liquid interface is continuously subjected to various perturbations. These perturbations can appear either as white-noise-type random fluctuations or as colored noise with spectral selectivity. The former corresponds to atomic thermal motion and unresolved background disturbances under experimental conditions. It is widely employed in numerical simulations as an approximation of thermal fluctuations and has successfully reproduced experimental phenomena such as dendritic side-branching and interfacial instability. In contrast, the latter frequently originates from external fields or fluid-driven effects, such as thermal flux fluctuations, ultrasonic vibrations, and electromagnetic pulses. Such external forces can amplify interfacial dynamics within specific frequency ranges. For instance, Fetzer et al. [18] showed through thin liquid film dewetting experiments that thermal fluctuations can alter the interfacial fluctuation spectrum within certain wavenumber ranges. Similarly, Liotti et al. [19] used synchrotron imaging to reveal the spectral effects of pulsed electromagnetic fields on dendrite fragmentation rates and interfacial dynamics. Zhang et al. [20] demonstrated through tomographic imaging under ultrasonic excitation that mechanical vibrations influence dendritic restructuring according to their frequency modes. Furthermore, Neumann-Heyme et al. [21] conducted a quantitative investigation of dendrite tip morphological responses under various perturbation conditions using high-resolution in situ imaging. These experimental findings indicate that perturbations during solidification involve not only a nonselective white-noise background but also colored noise with distinct spectral characteristics.

Theoretical research also highlights the ubiquity of colored noise in non-equilibrium systems. Lisy and Tothova [22] systematically analyzed the relevant noise mechanisms within transport theory, demonstrating their widespread presence in statistical physics. Jou and Galenko [23], working within the framework of non-equilibrium thermodynamics, showed that noise characteristics can significantly influence solidification pattern selection and the evolution of interfacial dynamics. Tegze et al. [24] introduced noise into the PFC model, which was truncated at high wavenumbers to suppress nonphysical atomic-scale fluctuations and improve numerical stability. Such noise is effectively short-ranged and does not exhibit spatial correlations over mesoscopic length scales. Consequently, it does not lead to spectral mode selection or interfacial destabilization and was not intended to describe noise-driven morphological instabilities. However, in the PFC framework, noise acts directly on the atomic-scale density field, manifesting only as local perturbations at the lattice scale. As a result, it fails to capture interface destabilization driven by mesoscopic or macroscopic perturbations [24]. In the APFC model, noise acts on the coarse-grained complex amplitudes, modulating the envelope intensity and phase of the lattice. This mechanism enables the excitation of large-scale interfacial perturbations, leading to morphological transformations such as tip splitting and seaweed-like growth. In contrast to conventional phase-field models, which lack atomic-scale resolution and therefore cannot account for wavelength-dependent or colored noise effects, the APFC framework provides an effective means of investigating the influence of spectrally selective noise on crystal growth.

Building on these findings, this study systematically investigates the effects of Gaussian colored noise on dendritic growth within the APFC framework, with particular emphasis on how its intensity, filtering width, and characteristic wavenumber regulate tip dynamics and interfacial stability. The aim is to clarify the mechanisms by which different types of noise influence the evolution of dendritic morphology, thereby providing theoretical guidance for optimizing solidification microstructures through external fields or processing-induced perturbations.

2. APFC model and simulation methods

2.1. Amplitude-expanded of the Phase Field Crystal Model

The original dimensionless phase field crystal (PFC) model is based on the Swift–Hohenberg (SH) model, with the free energy functional defined as [7]:

$$F = \int d\mathbf{r}\left[\frac{a_0}{2}(\nabla^2 + q_0)^2\psi + \frac{a_2}{2}\psi^2 + \frac{a_3}{3}\psi^3 + \frac{a_4}{4}\psi^4\right] \tag{1}$$

Here, the parameters a_0, a_2, a_3, a_4 are dimensionless model quantities determined by factors such as degree of undercooling, the isothermal compressibility of the liquid phase, and the elastic constants of the crystal. These parameters can be quantitatively fitted using experimental measurements or molecular dynamics data [25].

Specifically, the gradient coefficient a_0 is related to the elastic properties of the crystal and the interface energy, expressed as:

$$a_0 = \frac{k_B T \Gamma}{\rho_0 q_0^4}, \tag{2}$$

where k_B is the Boltzmann constant, T is the temperature, ρ_0 is the reference liquid phase density, q_0 is the principal wave vector of the structural factor, and Γ is the curvature coefficient of the structural factor convolution kernel.

The parameter a_2 correlates with the degree of undercooling, given by

$$a_2 = -\frac{k_B T}{\rho_0}\left(\frac{1}{S(q_0)} - \frac{b_1^2}{4b_2}\right), \tag{3}$$

where $S(q_0)$ denotes the structural factor peak and b_1, b_2 are the coefficients of nonlinear terms in the local free energy [25]. These coefficients can be obtained by fitting the steady-state density wave amplitude:

$$b_1 = \frac{3}{2S(q_0)\mu_s}, \quad b_2 = \frac{4}{30S(q_0)\mu_s^2}. \tag{4}$$

Where μ_s denotes the equilibrium amplitude of the crystalline density modulation in the bulk solid phase, obtained by minimizing the bulk free-energy density. The cubic term coefficient a_3 describes the asymmetry of the solid–liquid phase free energy well; when this asymmetry is negligible, a_3 can be set to zero. The quartic coefficient a_4

controls the nonlinear saturation of the density wave, expressed as [25]:

$$a_4 = \frac{k_B T_b}{3\rho_0^3}. \tag{5}$$

In summary, all parameters of the PFC model can be quantitatively fitted from experimentally measured quantities such as the structure factor, density wave amplitude, and liquid phase density, thereby establishing a direct link between the physical properties of real materials and the theoretical model.

The dimensionless atomic number density field ψ exhibits periodic variation reflecting the atomic arrangement in the crystal structure. It can be approximated through a complex amplitude expansion:

$$\psi(\mathbf{x},t) = \bar{\psi}(\mathbf{x},t) + \sum_{j=1}^{3}\left[\eta_j(\mathbf{x},t)\exp\left(i\mathbf{q}_j\cdot\mathbf{x}\right) + \eta_j^*(\mathbf{x},t)\exp\left(-i\mathbf{q}_j\cdot\mathbf{x}\right)\right] \tag{6}$$

Here, $\bar{\psi}$ represents the average density field, which determines the driving force for crystal growth. η_j and η_j^* are the complex amplitudes associated with reciprocal lattice vector $\mathbf{q}_j$ and its conjugate. In this study, we consider only the two-dimensional triangular lattice, which has three reciprocal lattice vectors:

$$\mathbf{q}_1 = \frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j}, \mathbf{q}_2 = \hat{j}, \mathbf{q}_3 = -\frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j}. \tag{7}$$

where $\hat{i}$ and $\hat{j}$ are the unit vectors in the x and y directions, respectively. Using the variational method, Dong-Hee Yeon et al. [12] proposed a complex amplitude expansion of the crystal phase field (APFC) model based on the PFC model. In this model, the complex amplitude fields η_j and the average density field $\bar{\psi}$ describe the evolution of the microstructure in the system.

2.2. Gaussian colored noise

To investigate the effect of noise on crystal growth a Gaussian colored noise term $\zeta(\mathbf{x},t)$ is introduced into the complex amplitude evolution equation:

$$\begin{aligned}\frac{\partial\bar{\psi}}{\partial t} = \nabla^2\Big[&(a_0+a_2)\bar{\psi} + a_2\bar{\psi}^2 + a_4\bar{\psi}^3 + 2(a_3+3a_4\bar{\psi})\sum_{i=1}^{3}|\eta_i|^2 \\ &+ 6a_4\left(\prod_{i=1}^{3}\eta_i + \prod_{i=1}^{3}\eta_i^*\right)\Big],\end{aligned} \tag{8}$$

and

$$\begin{aligned}\frac{\partial\eta_j}{\partial t} = -\Big\{&a_0(\nabla^2 + 2i\mathbf{q}_j\cdot\nabla)^2\eta_j + \left(a_2 + 2a_3\bar{\psi} + 3a_4\bar{\psi}^2\right)\eta_j \\ &+ 2(a_3+3a_4\bar{\psi})\prod_{i\neq j}^{3}\eta_i^* + 3a_4\eta_j\left(2\sum_{i=1}^{3}|\eta_i|^2 - |\eta_j|^2\right)\Big\} + \zeta.\end{aligned} \tag{9}$$

The evolution Eqs. (8) and (9) are numerically solved using a semi-implicit Fourier spectral method, which efficiently handles the nonlinear terms while maintaining high numerical stability.

The statistical properties of this noise are given by:

$$\langle\zeta(\mathbf{x},t)\zeta(\mathbf{x}',t')\rangle = -D\nabla^2 g(|\mathbf{x}-\mathbf{x}'|,\lambda)\delta(t-t'). \tag{10}$$

Here, $g(|\mathbf{x}-\mathbf{x}'|,\lambda)$ characterizes the spatial correlation of the noise in real space. The noise intensity D controls the overall perturbation amplitude, and $\delta(t-t')$ is the temporal Dirac delta function, describing the temporal correlation of the noise [26]. In Fourier space, the colored noise is generated using an explicitly real and symmetric spectral filter of the form

$$\hat{g}(k) = \exp\left[-\frac{(k-k_{peak})^2}{2\lambda^2}\right] \tag{11}$$

Where λ characterizes the spectral width of the noise, and k_{peak} denotes the characteristic wave number at which noise power is concentrated. Together, these parameters define the spectral distribution of the noise. Gaussian colored noise allows selective amplification of perturbations at specific wave numbers, making it possible to analyze how different spectral components of noise contribute unequally to interfacial instability, which cannot be resolved using white noise.

2.3. Numerical simulation settings

Simulations are conducted with two-dimensional periodic boundary conditions. Initial conditions include small crystal nuclei embedded in the liquid and a flat solid–liquid interface. The evolution equations are integrated numerically using a semi-implicit Fourier spectral method to ensure both stability and efficiency. Self-developed GPU code is employed for computation, yielding efficient and stable results.

The simulated solid–liquid system has dimensions of $650a \times 650a$ (with a the lattice constant), discretized with grid spacing $dx = dy = 0.783$, and a time step of $dt = 0.1$. The large system size minimizes boundary reflection artifacts, thereby ensuring the reliable statistics for dendritic growth morphologies. For dendrite simulations, the initial condition was set as a circular nucleus with a radius of $10a$, positioned at the center of the simulation box.

In this study, the parameters are chosen as $a_0 = 1$, $a_3 = 0$, and $a_4 = 1$, corresponding to the classical SH model. In two-dimensional systems, the APFC model is capable of describing the triangular phase, striped phase, and liquid phase, as well as the liquid–triangular and triangular–striped two-phase coexistence regions [8]. During crystal growth, the evolution of crystal morphology is governed by both kinetic constraints and interface energy. The competition between kinetic effects and interfacial energy gives rise to different growth modes, which are associated with crystal anisotropy (or symmetry) and the degree of undercooling, respectively. Within the phase-field crystal framework, both the crystal symmetry and the degree of undercooling are controlled by the model parameters a_2 and $\bar{\psi}$. In this study, we focus on the triangular–liquid coexistence region and investigate the solidification behavior of triangular crystals under different degrees of undercooling by varying a_2 and $\bar{\psi}$, with particular emphasis on the effects of Gaussian colored noise.

3. Results and discussion

3.1. Evolution of dendritic morphology under noise-free conditions

Before examining the effects of noise, we first systematically investigate crystal growth under various undercooling conditions in a noise-free environment, thereby establishing a baseline for subsequent comparisons. The two key parameters in this study are the undercooling coefficient a_2 and the average density $\bar{\psi}$. In the APFC model, a_2 is a constant proportional to the dimensionless undercooling and corresponds to interface energy anisotropy, thereby influencing dendrite orientation selection and morphological stability. In contrast, $\bar{\psi}$ controls the free energy difference between the liquid and solid phases, determining the relative stability of the solid–liquid coexistence state. By jointly adjusting a_2 and $\bar{\psi}$, a wide range of growth conditions can be explored, spanning from low driving forces with weak instabilities to high driving forces with strong instabilities.

The distribution of the ordinate parameter field $\boldsymbol{\Phi}$ under different $(a_2,\bar{\psi})$ conditions is shown in Fig. 1, where $\boldsymbol{\Phi} = 2\sum_{j=1}^{3}\eta_j\eta_j^*$. The results clearly depict the evolution of crystal growth under varying parameter conditions $(a_2,\bar{\psi})$. The selected parameters, based on the APFC phase diagram, indicate that variations in growth driving force and crystal anisotropy lead to significant changes in crystal morphology. The phase diagram in Fig. 1(o), which provides an enlarged view of the liquid phase, triangular phase, and their coexistence region, is constructed based on the one-mode approximation and the Maxwell equal-area construction [8]. This diagram serves as a basis for selecting parameter sets $(a_2,\bar{\psi})$ used in the subsequent simulations. The evolutionary patterns under different parameter conditions can be summarized as follows:

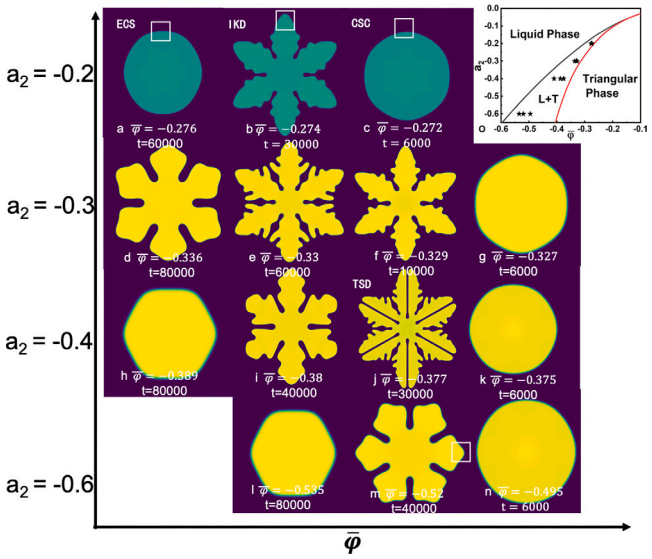


Fig. 1. Crystal growth morphologies across solid–liquid coexistence conditions without noise. The inset (upper right) marks parameter locations within the solid–liquid coexistence region, where decreasing values of $(a_2, \bar{\psi})$ in the phase diagram correspond to stronger interface energy anisotropy and higher driving force.

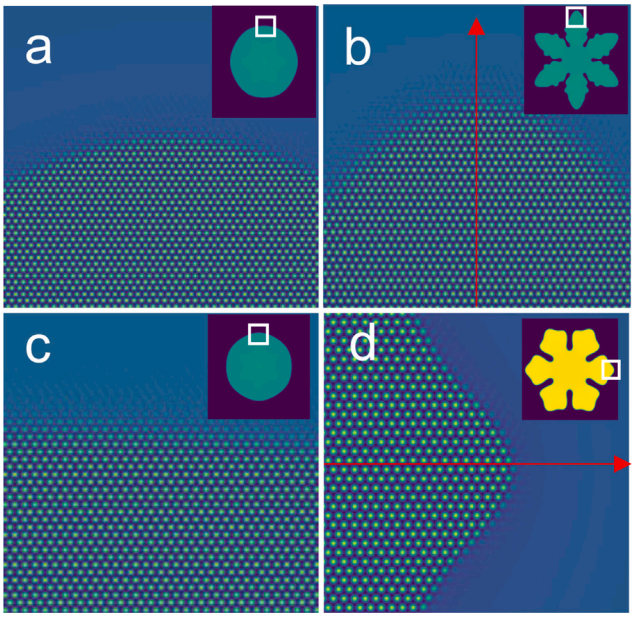


Fig. 2. Atomic arrangements at dendrite tips for typical morphologies in Fig. 1: (a) equilibrium crystal shape (ECS), (b) interface kinetic dendrite (IKD), (c) compact sphere-shaped crystal (CSC), and (d) interface energy dendrite (IED). The main arm orientations in (b) and (d) align with open-packed and close-packed lattice directions, respectively, confirming that APFC captures lattice orientation effects.

Compact sphere-shaped crystal growth under high drive force: When $a_2 = -0.2$ and the average density approaches the triangular phase, for example $\bar{\psi} = -0.272$, the degree of undercooling is extremely high. Under this condition, the crystal growth rate becomes faster, and the morphology exhibits an isotropic, circular compact sphere-shaped crystal (CSC) (see Fig. 1(c)). This phenomenon can be attributed to the higher growth driving force under high-density conditions, which suppresses interfacial instability and consequently promotes isotropic growth.

Evolution from hexagonal dendrites to equilibrium crystal shape: As $\bar{\psi}$ decreases (Fig. 1(b)), the degree of undercooling is rapidly reduced, and crystal growth transitions to hexagonal dendrites. At this stage, the crystal morphology becomes more influenced by the diffusion process, as interface energy plays a diminished role in the growth dynamics. This type of dendrite is referred to as an interface kinetic dendrite (IKD). The growth direction and speed of the dendrites are primarily governed by diffusion-driven kinetic constraints at the interface. However, due to the weaker influence of interface energy, the dendritic structures are relatively simple, often exhibiting more symmetrical shapes with smoother boundaries. The primary growth direction of the dendrite tips aligns with the crystallographic open-packed direction, corresponding to the orientation of maximum surface energy, as seen in Fig. 2(b) [27]. As the density decreases further, growth becomes dominated by interface energy, resulting in a very slow growth rate and the formation of equiaxed crystals (ECS) with the atomic arrangement illustrated in Fig. 2(a).

Kinetics-driven formation of dendrites: When $a_2 = -0.3$, as the average density decreases, the degree of undercooling diminishes gradually. The dendrite arms gradually become elongated and begin to form primary branches. Under moderate undercooling, strong coupling between interfacial energy and diffusion kinetics amplifies interfacial instability, leading to relatively short primary branches. As the density decreases further, the weakening of kinetic driving forces causes interfacial energy to dominate crystal growth, leading to a more regular grain morphology and a corresponding reduction in the scale of secondary dendrites.

Evolution of tip-splitting dendrites: As a_2 is further reduced to −0.4, with increasing average density, undercooling increases more gradually, and the region of strong coupling between kinetics and interface energy anisotropy enlarges, resulting in crystal growth that exhibits a more complex tip-splitting dendrite (TSD) pattern (see Fig. 1(j)). TSD is a distinct dendritic growth mode primarily governed by the interplay between interface energy anisotropy and kinetic anisotropy. Under moderate undercooling conditions, this intensified competition between interface energy anisotropy and kinetic anisotropy causes the crystal growth to lose the constraint of a single preferential orientation, ultimately resulting in the development of intricate dendritic structures. This behavior is consistent with the Mullins–Sekerka instability theory [27,28]. With decreasing undercooling, the grain morphology reverts to hexagonal dendrites as observed for other a_2 values, where interface energy anisotropy becomes the dominant factor.

Hexagonal symmetric dendrites dominated by interface energy anisotropy: When a_2 is further reduced to −0.6, the degree of undercooling changes very slowly with average density. At the three different equilibrium density values explored in this study, undercooling is minimal, and interface energy anisotropy dominates the preferential growth direction of dendrite tips, yielding a perfectly sixfold symmetric interface energy dendrite (IED) morphology (see Fig. 1(m)). According to the atomic arrangement (see Fig. 2(d)), it can be observed that the primary arms of the dendrite align with the close-packed direction of the crystal. This fundamental reason for this directional selection lies in the inherent interface energy anisotropy of the crystal structure, which guides the dendrites to preferentially grow along the directions with the lowest energy. This IED pattern closely matches surface tension-driven dendrite growth patterns observed in Hele-Shaw experiments [27].

Fig. 2 displays the atomic arrangements at different dendrite tips, corresponding to ECS, IKD, CSC, and IED, respectively. It is clearly visible that the dendrite main arm in Fig. 2(b) tends to extend along the lattice's open-packed direction, consistent with orientation selection dominated by interface kinetics. In contrast, the dendrite main arms in Fig. 2(d) align with the close-packed direction of the crystal lattice, indicating that interface energy anisotropy dominates in this case. This validates the physical fidelity of the APFC for complex solidification

interface evolution and underscores its utility for studying dendritic pattern selection under varying driving forces and anisotropies.

In summary, the APFC model effectively reproduces various typical dendrite growth patterns reported in experiments and PFC simulations [8,9], encompassing the systematic evolution from CSC, ECS, IED, IKD to TSD. This validates the physical fidelity of the APFC for complex solidification interface evolution and underscores its utility for studying dendritic pattern selection under varying driving forces and anisotropies.

3.2. Growth dynamics and morphological transition mechanisms

To quantify growth kinetics under different undercooling conditions, we computed the temporal evolution of the dendrite tip velocity. The time derivative of the tip position $Xtip(t)$ defines the instantaneous tip velocity $V(t)$. As shown in Fig. 3(a), all operating conditions display consistent evolutionary characteristics. The process begins with an initial rapid growth phase, characterized by short-term acceleration immediately after nucleation, primarily driven by density and interfacial curvature. Subsequently, as the local density decreases or the undercooling varies, the tip velocity gradually declines and eventually stabilizes at a steady state. This temporal pattern provides a unified reference along a common time axis for correlating velocity and morphology across different dendritic types (CSC, ECS, IED, IKD, and TSD).

In noise-free conditions, the steady tip velocity increases monotonically with average density, and morphology maps to velocity interval rather than a one-to-one correspondence.

Low-speed zone (ECS): Under low undercooling or limited effective diffusion, the interface remains nearly isotropic, giving rise to the ECS. At this stage, side branches are absent and the interface roughness is minimal (see Figs. 1(a) and 2(a)).

Intermediate-speed zone (IED → TSD): As the undercooling increases, interface energy anisotropy governs tip orientation producing hexagonal dendrites (IED). With accumulating kinetic anisotropy and diffusion shielding, tip splitting appears, and the system enters the TSD state, characterized by higher roughness and dense lateral branching (Fig. 1(j)).

High-speed zone (IKD → CSC): At high undercooling, interface kinetics dominate, and the driving force acts directly at the front. When suppression is weak, rapid local atomic rearrangement thins the tip and produces short, dense side branches, resulting in kinetically-driven dendrites (IKD). With still stronger driving forces, interface undulations are effectively suppressed, the front becomes smooth and stable, and growth approaches an isotropic CSC.

The velocity-morphology correspondence is consistent with the systematic evolution obtained by jointly tuning a_2 and $\bar{\psi}$ described in Section 3.1: decreasing $|\bar{\psi}|$ increases the undercooling, drives the system deeper into the solid–liquid coexistence region, and a concurrent reduction of $|a_2|$ promotes the gradual transition IED → TSD → IKD.

The underlying mechanism is the competition between interface energy anisotropy and interface kinetic anisotropy. When interface energy anisotropy dominates, the main arm advances steadily along the close-packed lattice direction, giving a highly symmetric IED morphology. As interface kinetic effects strengthen, diffusion shielding intensifies and the tip curvature radius decreases. Perturbations within the instability spectrum are preferentially amplified at the tip, triggering tip splitting and the transition to TSD. With further increase in the driving force, interfacial migration becomes kinetically controlled; side-branch growth is constrained to short, dense structures, thereby leading to IKD.

The amplitude approximation of APFC suppresses non-physical high-frequency oscillations while preserving lattice orientation effects. Consequently, the tip velocity time curve exhibits smoother than those in conventional PFC simulations, yet the critical morphology criteria and sequence remain consistent, corresponding to the steady-state plateaus under different growth conditions (Fig. 3(a)).

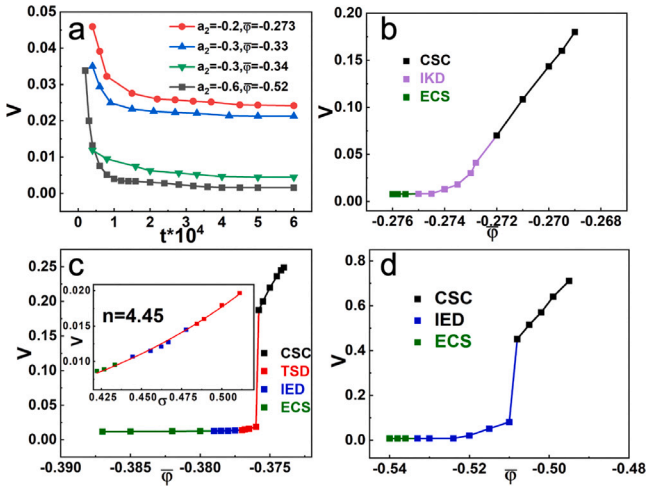


Fig. 3. (a) Temporal evolution of dendrite tip velocity V under different growth conditions. (b–d) Dendrite tip velocity V as a function of the average density $\bar{\psi}$. The inset in (c) presents V as a function of the supersaturation degree σ.

The curves of dendrite tip velocity (V) as a function of the average density $\bar{\psi}$ are shown in Fig. 3(b–d). The results show that the tip velocity increases monotonically with $\bar{\psi}$. In the APFC model, increasing the average density effectively reproduces the experimental enhancement of the growth driving force, which is physically equivalent to raising the supersaturation degree [8].

To further delineate the boundary between the kinetics-dominated and diffusion-dominated regimes, we analyzed the scaling relationship between the steady-state tip velocity and the driving force, represented by the equivalent supersaturation, $\sigma = \frac{\bar{\psi}_0 - \bar{\psi}_L^e}{\bar{\psi}_S^e - \bar{\psi}_L^e}$, $\bar{\psi}_L^e$ and $\bar{\psi}_S^e$. Here, the $\bar{\psi}_L^e$ and $\bar{\psi}_S^e$ represent the liquid phase and solid phase lines, respectively, within the two-dimensional hexagonal-liquid coexistence region of the phase diagram, as shown in Fig. 3(c).

First, the morphologies corresponding to ECS, IED, and TSD conform to typical diffusion-controlled scaling laws. Fitting results reveal a clear power-law relationship ($V = A\sigma^n$) between the steady-state tip velocity and the driving force, with an exponent of approximately $n \approx 4.45$. This satisfies the expectation that $n > 2$ under diffusion-controlled conditions. The observation agrees with the pattern predicted by the tip-curvature-solute-supply-diffusion-field coupling mechanism within the Lipton–Kurz–Trivedi (LKT) theoretical framework [29].

By contrast, the simulation results for CSC and certain IKD conditions deviate significantly from this power-law relationship, instead exhibiting an approximately linear trend. This behavior indicates that, under these conditions, interface kinetics rather than diffusion processes dominate the determination of tip velocity.

In conclusion, the results demonstrate that crystal growth in ECS, IED, and TSD is predominantly governed by the diffusion mechanism, while the growth modes of IKD and CSC lie between diffusion-controlled and interface-kinetics-controlled regimes, reflecting a more complex growth mechanism. Compared to the original PFC model [8,9], the APFC model can similarly capture different types of dendritic growth, with growth rate curves exhibiting an overall trend comparable to PFC. However, the APFC growth rate curves are significantly smoother. This smoothing primarily arises from the amplitude expansion approximation: APFC effectively smooths the interface migration process, removing the lattice pinning effect present in the original PFC under low driving forces or rate-limiting conditions [30,31]. As a result, the interface reaction is underestimated, and the velocity curve becomes excessively smoothed.

Nevertheless, the APFC model remains capable of effectively establishing quantitative relationships between growth parameters, crystal

morphology, and average velocity, while accurately capturing the overall evolution trend at the interface. We calculated the interfacial energy based on

$$\gamma = \frac{1}{L}\int dr\left[f(r) - \left(f_s\frac{\bar{\psi}-\bar{\psi}_l}{\bar{\psi}_s-\bar{\psi}_l} - f_l\frac{\bar{\psi}-\bar{\psi}_s}{\bar{\psi}_s-\bar{\psi}_l}\right)\right]. \tag{12}$$

Where $f(r)$ is the free energy density of the simulated system, $\bar{\psi}_l$ and $\bar{\psi}_s$ denote the average densities of the liquid and solid phases, respectively, f_l and f_s are the corresponding average free energies, and L is the total length of the interface [32,33]. For the parameter $a_2 = -0.3$, we obtain an interfacial energy of $\gamma = 1.15$, which is in good agreement with values widely reported for the original PFC model in the literature [33]. Furthermore, the kinetic coefficient extracted from the relation $Z = Z_0 + C(t+t_0)^{1/2}$ [33], yields a value of approximately 1.43, again consistent with previously reported PFC-based results [33]. These agreements indicate that, despite its coarse-grained nature, the APFC model captures key quantitative features of interfacial energetics and kinetics.

3.3. Effect of gaussian colored noise on crystal growth

Sections 3.1 and 3.2 on dendritic morphology and growth mechanisms provide the theoretical foundation for understanding dendritic evolution. However, during solidification, the solid–liquid interface is inevitably subjected to perturbations arising from intrinsic thermal fluctuations or external influences, such as mechanical vibrations and fluid disturbances. These perturbations exhibit distinct spectral distributions, with Gaussian colored noise as a prominent example. Prior studies have shown that such noise, characterized by spectral correlations, significantly influences interfacial stability and dendritic evolution, particularly under conditions of high undercooling and strong driving forces [24,34]. A more thorough understanding of its role is thus essential for elucidating the microscopic mechanisms governing solidification.

To achieve this issue, we begin with the simplest structural model — a planar solid–liquid interface, as the focus of study. Within this framework, the effects of characteristic wavenumber on interfacial stability are systematically examined, revealing a selective coupling between the noise spectrum and instability modes. Analysis of the planar interface yields both fundamental insights and quantitative relationships, establishing a foundation for further investigation of noise mechanisms. Building on this foundation, circular nucleation interfaces are then introduced to better represent the nucleation and growth conditions encountered in practical solidification processes. This approach makes it possible to elucidate how noise influences dendritic growth in complex geometries. By comparing with noise-free conditions, we demonstrate how noise amplifies tip perturbations in intricate interface morphologies, resulting in complex dendritic structures.

By comparing planar and circular nucleation interfaces, this section examines the specific effects of noise intensity, filter width, and characteristic wavenumber on dendritic morphology evolution. Incorporating insights from experimental observations, this analysis aims to comprehensively elucidate the role of noise in interfacial stability and growth kinetics during solidification.

3.3.1. Spectral selectivity and interfacial instability at the planar solid–liquid interface

As an idealized model, the planar interface eliminates geometric complexity, enabling a focused analysis of the interactions between noise and interfacial instability modes. The methodology consists of deriving dispersion relations through linear stability analysis to determine the stability range for the interface under varying conditions, followed by numerical simulations to investigate evolution patterns under different characteristic wavenumbers. This combined approach links theoretical predictions with numerical results and provides a comprehensive physical understanding.

Dispersion relation and instability criterion. To elucidate the physical mechanisms governing the influence of noise on dendritic morphology, we first analyze the dispersion relation $\alpha(k)$, which characterizes the linear growth rate under various parameter conditions. In accordance with the PFC approach, a comparable linearization of the complex amplitude field is performed [35]:

$$\eta(\mathbf{x},t) = \eta_{\text{steady}} + \delta\eta(\mathbf{x},t). \tag{13}$$

Here, η_{steady} denotes the complex amplitude of the steady-state field. A Fourier transform is then applied to the perturbation term, converting the governing equation from real space to Fourier space:

$$\delta\eta = A\exp(\omega t + ik\mathbf{x}) + C.C.. \tag{14}$$

Here, $\omega(k)$ denotes the complex growth rate, whose real part $Re(\omega(k)) = \alpha(k)$ represents the linear growth rate of perturbations with wave number k. By substituting these perturbation terms into the evolution Eq. (9) and neglecting all nonlinear terms of second order and higher, the linear growth rate of perturbations is obtained:

$$\alpha(k) = -a_0(k^2 + 2\mathbf{q}_j k)^2 - (a_2 + 2a_3\bar{\psi} + 3a_4\bar{\psi}^2), \tag{15}$$

where $\alpha(k)$ represents the linear growth rate of perturbations with wave number k in the system.

It is important to emphasize that this derivation is independent of spatial dimensionality. In one-dimensional systems, the wave vector k reduces to a scalar, with $\mathbf{q}_j$ defaulting to unity. In two-dimensional systems, where $k = (k_x, k_y)$, the density field expansion involves multiple sets of reciprocal lattice vectors, and the linear operator acting on perturbations along different $\mathbf{q}_j$ directions exhibits slight variations. When $\alpha(k) > 0$, the corresponding perturbation grows exponentially in time, rendering the system linearly unstable. When $\alpha(k) < 0$, the perturbation decays exponentially, and the system remains linearly stable or metastable. Differentiating $\alpha(k)$ with respect to k gives $\alpha'(k)$. Since $a_0 > 0$, for $k > 0$, $\alpha'(k) < 0$ indicates that $\alpha(k)$) decreases monotonically, attaining its maximum at $k = 0$.

Subsequent analysis shows that, when other parameters held constant, an increase in average density $\bar{\psi}$ results in a corresponding rise in the overall system stability parameter. The parameter $\alpha(k)$ may shift from the negative to the positive domain, thereby enhancing the system instability. As illustrated in Fig. 4(a), under shallow undercooling (e.g., $a_2 = -0.3$, $\bar{\psi} = -0.33$), $\alpha(k) < 0$ for all k, and the system remains in a linearly stable metastable state, requiring external perturbations to overcome the energy barrier for phase transitions. When the average density increases (e.g., $a_2 = -0.3$, $\bar{\psi} = -0.31$), the dispersion curve enters the positive domain, indicating that the system has transitioned into an unstable regime where even small perturbations can grow exponentially. In this case, if the dominant noise wavenumber lies within the $\alpha(k) > 0$ interval, the interface becomes highly susceptible to instability. By contrast, when the perturbation wavenumber falls within the $\alpha(k) < 0$ interval, the influence of noise remains relatively weak.

Extending the analysis to two dimensions, we plotted the distribution of $\alpha(k_x, k_y)$ under the conditions $a_2 = -0.3$ and $\bar{\psi} = -0.31$. The results show that in a hexagonally symmetric system, $\alpha(k_x, k_y)$ exhibits symmetric maxima along the three equivalent reciprocal lattice vector directions. Numerically, the maximum peak reaches approximately 0.0117, with corresponding wave-vectors $(k_x, k_y) \approx (-1.76, 0.06)$, $(-1.00, -1.04)$, and $(1.76, 0.06)$, which align with the reciprocal lattice vectors $\mathbf{q}_1 = \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$, $\mathbf{q}_2 = (0, 1)$, and $\mathbf{q}_3 = \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$, respectively. These values provide theoretical benchmarks for subsequent analyses of interface perturbations.

Then, numerical experiments were carried out under the conditions $a_2 = -0.3$ and $\bar{\psi} = -0.31$ to further investigate the influence of different noise parameters on interfacial stability.

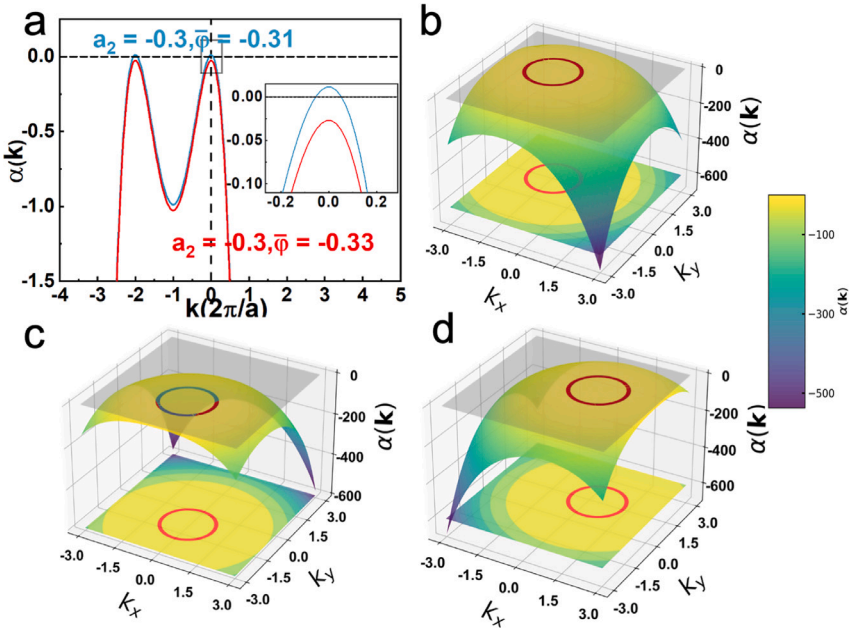


Fig. 4. Dispersion relations. (a) One-dimensional dispersion relations under two different parameter sets. (b–d) Two-dimensional dispersion relations and their projections along the reciprocal lattice vector directions. $\mathbf{q}_1 = \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$, $\mathbf{q}_2 = (0, 1)$, and $\mathbf{q}_3 = \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$. Red regions indicate $\alpha(k_x, k_y) > 0$, while the gray plane denotes $\alpha(k_x, k_y) = 0$. Parameters: $a_2 = -0.3$, $\bar{\psi} = -0.31$.

Effect of characteristic wavenumber. To more clearly elucidate the role of noise in triggering and regulating planar interface instabilities, we performed an additional set of simulations in which a small random perturbation was introduced only at $t = 0$, while no dynamic noise was applied thereafter ($D = 0$). When the perturbation wave vector is chosen along the grid direction (0,1) with $k_x = 0$, the problem effectively reduces to one dimension, and the numerical results are fully consistent with the analytical dispersion relation. This configuration was therefore selected as a representative case to examine the intrinsic linear instability of the planar interface. We therefore selected this direction as a representative case to systematically investigate the influence of noise on planar interface instabilities. Fig. 5(a) shows the evolution in the absence of any initial perturbation and with a constant dynamic noise intensity $D \equiv 0$ throughout the simulation. Under these idealized conditions, the interface remains perfectly flat over the entire accessible simulation time. By contrast, when noise is introduced with a vertical characteristic wave number $k_{peak} = 0$ (corresponding to the maximum perturbation wave vector), noise intensity $D = 0.05$, and filter width $\lambda = \lambda_0 = 0.02(2\pi/a)$, yielding $\alpha(k) = 0.0117$, small initial interfacial fluctuations (Fig. 5(b)) are progressively amplified. These perturbations eventually develop into a seaweed-like morphology (Fig. 5(c)), with the dominant interfacial wave number extracted from the simulations being $k_{pattern} = 0.0262$. This behavior is consistent with the classical Mullins–Sekerka instability theory [36,37]. It is noteworthy that, because the noise directly perturbs the complex amplitude variable rather than the atomic density field, such disturbances are more effectively amplified at the mesoscale. This manifests as interfacial morphological changes and produces dendritic evolution patterns similar to those predicted by the phase field theory [38]. These results demonstrate that spatially correlated noise introduced into the complex amplitude field is more effective in triggering macroscopically observable nonlinear interfacial dynamics than perturbations applied directly to the PFC density field.

When colored noise with $k_{peak} = 0.2$ is introduced, $\alpha(k) = -0.1819$, leading only to minor interface fluctuations without the development of pronounced dendritic morphologies. The corresponding fluctuation wave number is 0.0215. For a larger noise peak at $k_{peak} = 0.5$ ($\alpha(k) = -1.5508$), interface undulations are further suppressed, resulting in a nearly flat interface with a characteristic wave number of 0.0077. These results demonstrate that the choice of noise vector directly determines whether linear-stage interface instabilities can be triggered. As illustrated in Fig. 5(d), the final interface pattern is strictly controlled by both the sign of the dispersion relation $\alpha(k)$ and the location of the noise spectral peak. As k_{peak} increases, the corresponding $\alpha(k)$ rapidly decreases; when $\alpha(k)$ is significantly negative, interface roughening is strongly inhibited, and $k_{pattern}$ approaches zero. This comparison demonstrates that dendritic morphology does not result from the indiscriminate noise amplification but is instead governed by the unstable wavenumber range defined by the system's dispersion relation $\alpha(k)$. Tang et al. [39] further showed that low-frequency electromagnetic vibrations enhance solute fluctuations, thereby promoting dendrite fragmentation and facilitating the transition from columnar to equiaxed crystals. These findings are consistent with the theory of selective noise amplification.

In crystalline systems, when the intrinsic instability is pronounced (i.e., when $\alpha(k)$ in the dispersion relation reaches higher values), noise wavenumbers within the unstable range are strongly amplified, thereby enhancing interfacial perturbations and accelerating morphological evolution. However, as the corresponding $\alpha(k)$ values decrease, the amplification effect of noise rapidly diminishes and may even be naturally suppressed by the system. This selective amplification mechanism indicates that the influence of noise is not uniform but is strictly governed by the unstable wavenumber range defined by $\alpha(k)$. Significant amplification of perturbations occurs only when the noise wavenumber coincides with this range, leading to pronounced nonlinear interfacial behavior. By contrast, in the lower $\alpha(k)$ regime, the impact of noise is negligible and insufficient to substantially alter the interface morphology. These results further highlight the APFC model's ability to capture noise-induced nonlinear interfacial dynamics with superior physical clarity and predictive capability during crystal growth.

It is important to emphasize that this process evolves slowly when only initial noise is applied, whereas the rate of interface morphology evolution increases markedly under continuous dynamic noise. With only initial noise, the interface exhibits only faint undulations at $t \approx 20,000$, much later than under dynamic noise, where similar

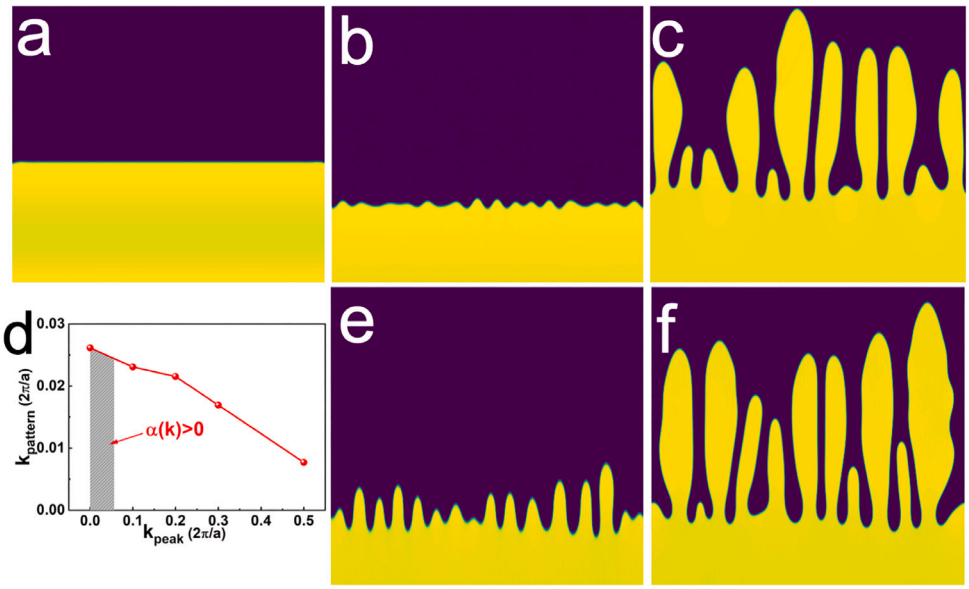


Fig. 5. Effect of noise parameters on planar interface dendrite growth. (a) Interface morphology in the absence of noise. (b,c) Interface morphologies at $t = 5000$ and $t = 20000$, respectively, for simulations with an initial perturbation characterized by a wave number of $k_{peak} = 0.1$. (d) Dependence of the dominant interface fluctuation wave number on the noise characteristic wave number for cases with initial perturbations. (e,f) Interface morphologies at $t = 1000$ and t=3000, respectively, for simulations with continuous dynamic perturbations characterized by a wavenumber $k_{peak} = 0.1$., Unless otherwise specified, the parameters are $a_2 = -0.3$, $\bar{\psi} = -0.31$, filter width $\lambda = 0.02(2\pi/a)$, and noise intensity $D = 0.05$.

features appear around $t \approx 3,000$ (Fig. 5(e, f)). In realistic solidification processes, thermal and concentration fluctuations are continuously present rather than as a one-time perturbation. Therefore, continuous dynamic noise more accurately reflects physical reality. In all subsequent spherical crystal growth simulations reported in this paper, the same continuous dynamic noise approach is adopted to efficiently trigger interface morphology evolution, yielding patterns consistent with experimental observations within a reasonable computational time.

In summary, the planar solid–liquid interface serves as a simple yet effective model system for elucidating the role of noise in crystal growth. The dispersion relation defines an intrinsic unstable wavenumber range, within which noise exhibits strong spectral selectivity. Collectively, these results demonstrate that the APFC model captures noise-induced nonlinear interfacial dynamics in a physically transparent manner and provides a robust theoretical basis for understanding the emergence of complex dendritic morphologies.

3.3.2. *Regulation of noise parameters and morphological evolution in circular nucleus interfaces*

Based on the analysis of noise-induced instability mechanisms in planar interfaces, we extended our investigation to a more complex circular grain growth system. This corresponds to the ground-state evolution under noise-free conditions shown in Fig. 1, enabling us to examine the regulatory effects of Gaussian colored noise with varying parameters on the actual morphological evolution. In these simulations, parameters $a_2 = -0.3$, $\bar{\psi} = -0.33$ were employed, differing from those adopted in planar interface studies. This adjustment was necessary because the previous planar conditions produced only CSC rather than dendritic morphologies. Compared to planar interfaces, circular nucleation interfaces more closely resemble realistic nucleation and growth scenarios during solidification. The inclusion of curvature effects and anisotropy in circular interfaces leads to more complex responses to noise.

Effect of characteristic wavenumber. The choice of noise wavenumber simultaneously determines the amplitude characteristics and spatial directionality of the perturbation. Similar to the previously described flat interface scenario, in numerical simulations of circular grain growth, colored noise with varying peak wavenumbers was introduced separately along the y-axis (Fig. 6(a)) and the x-axis (Fig. 6(b)) to systematically examine its effects on interface morphology evolution.

Taking the y-axis biased noise as an example, as shown in Fig. 6(a), when $k_{peak} = 0$, $\alpha(k) = -0.0267$, corresponding to the long-wavelength limit of the noise. In this case, the instability is strongest but lacks a distinct directional preference. Consequently, the liquid phase itself becomes highly unstable, leading to spontaneous solidification within the liquid, disruption of the interfacial structure, and the absence of well-defined dendritic morphology.

When $k_{peak} = 0.05$, $\alpha(k) = -0.0372$, dendritic growth begins to appear, with a relatively high number of dendrites. However, significant non-physical solidification still occurs in the liquid region, indicating that this wavenumber remains within the strongly unstable range.

When $k_{peak} = 0.1$, $\alpha(k) = -0.0708$, dendritic growth exhibits clear directional selectivity, with dendrites preferentially growing in the vertical direction, and the interface morphology responds markedly to noise perturbations.

When $k_{peak} = 0.2$, $\alpha(k) = -0.2203$, the noise perturbation falls into a relatively stable wavenumber range, and growth is significantly suppressed. The interface morphology responds weakly to noise, and the overall evolution tends toward stabilization.

For biased noise applied along the x-axis, the system shows an identical dependence on k_{peak} to that observed for the y-axis case, differing only in the orientation of the preferred dendritic growth.

In summary, the peak wavenumber of noise governs both the effective scale and amplification capacity of perturbations by selectively exciting linear modes across different wavenumber ranges, thereby determining the intensity of interface morphology responses. Smaller k_{peak} values preferentially excite unstable modes, where as larger k_{peak} values correspond to suppressed, more stable modes. When the characteristic noise wavenumber coincides with the region where the dispersion relation $\alpha(k)$ is large, perturbations are strongly amplified, driving interfacial instability. Conversely, directionally biased noise selectively enhances interface perturbations along specific orientations, producing pronounced anisotropy in morphology and growth behavior. This phenomenon arises from the anisotropic coupling between the amplitude field and perturbation wavevectors, indicating that the noise direction primarily dictates which interface modes or crystal orientations

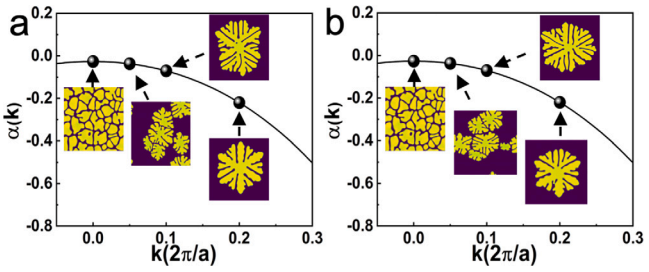


Fig. 6. The relationship between $\alpha(k)$ and wave number k. The four marked peak wave numbers of 0, 0.05, 0.1, and 0.2 correspond to the noise wave number parameters used in the numerical simulations of this paper. The insets show typical dendritic morphologies of circular grain growth under y-axis-biased noise (a) and x-axis-biased noise (b) at the selected peak wave numbers, with a filter width of $0.02(2\pi/a)$, noise intensity of 0.3, $a_2 = -0.3$, and $\bar{\psi} = -0.33$.

preferentially develop, without directly altering the overall instability strength of the system. In practice, factors such as applied temperature gradients, shear in flow fields, electromagnetic fields, or mechanical vibrations statistically introduce fluctuations with preferred orientations. Therefore, directionally biased noise more accurately represents the non-equilibrium perturbed environment experienced by interfaces in experimental systems. Moreover, perturbation wave vectors with selected characteristic length scales not only amplify interfacial instabilities but also impose directional selection on their spatial development, thereby promoting anisotropic crystal growth.

Effect of filter width. Filter width is also a key parameter for Gaussian colored noise, governing the frequency distribution — or the "color" of the noise, so that the noise wave numbers lie within the range $[k_{peak} - 2\lambda, k_{peak} + 2\lambda]$. Together, they define the spatial scale and temporal characteristics of perturbations, directly influencing both macroscopic and microscopic aspects of dendrite growth.

A narrow filter width ($\lambda = 0.8\lambda_0$, $\lambda_0 = 0.02(2\pi/a)$, $k_{peak} = 0.1$) corresponds to noise dominated by low-frequency components, characterized by large scales and slow perturbations (Fig. 7(a)). Under these conditions, dendrites retain a relatively regular morphology, with a thicker main stem and sparse side branches.

As shown in Fig. 7(b, c), with medium filter widths ($\lambda = \lambda_0$, $1.2\lambda_0$, $k_{peak} = 0.1$), the high-frequency components of noise gradually increase as the filter width broadens. The perturbation scale becomes smaller, the dendrite surface roughness increases, the density of side branches rises significantly, and the morphological complexity is enhanced.

At wide filter width ($\lambda = 2\lambda_0$, $k_{peak} = 0.1$), the noise contains a broad range of wave-number components, allowing perturbations across multiple length scales to be excited. Consequently, multiple nucleation points form in the liquid-phase region, from which dendritic growth emerges (Fig. 7(d)), resulting in a more complex branching structure.

In summary, the filter width controls the range of wave numbers present in the noise and, in the context of non-equilibrium solidification, plays a critical role by determining the number of unstable modes that can be excited, thereby influencing dendritic morphology and growth behavior. As the filter width increases, more wave-number components are introduced into the system, leading to higher side-branch density and increased morphological complexity. Optimizing both the characteristic wavenumber and the filter width to match regions of high linear growth rates is therefore essential for achieving precise control over dendritic morphology.

Effect of noise intensity on dendritic morphology. The noise intensity D is a key control parameter in Gaussian colored noise, representing the magnitude of random perturbations at the solid–liquid interface. Fig. 8 illustrates the evolution of dendritic morphology under varying noise intensities, with parameters $a_2 = -0.3$ and $\bar{\psi} = -0.33$.

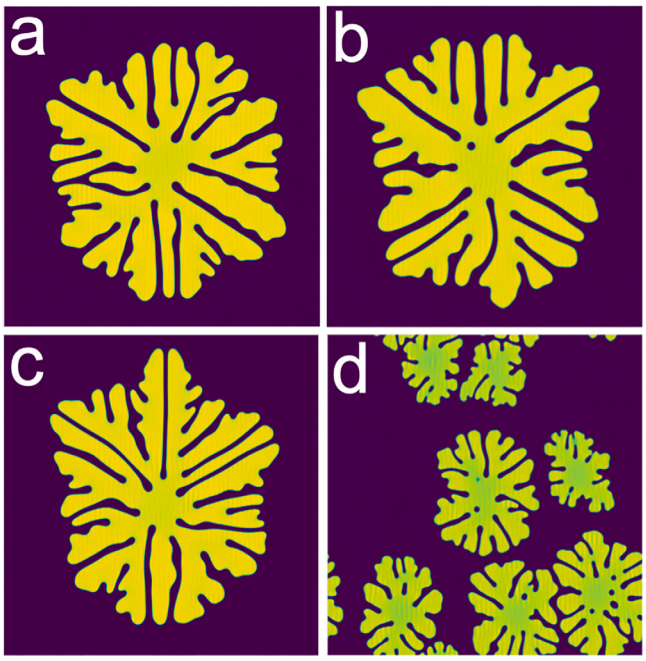


Fig. 7. Dendritic morphology at different filter widths for $a_2 = -0.3$, $\bar{\psi} = -0.33$. Panels (a)–(d) correspond to filter widths of $0.8\lambda_0$, λ_0, $1.2\lambda_0$, and $2\lambda_0$, respectively, with $\lambda_0 = 0.02(2\pi/a)$, noise intensity $D = 0.3$, and $k_{peak} = 0.1$.

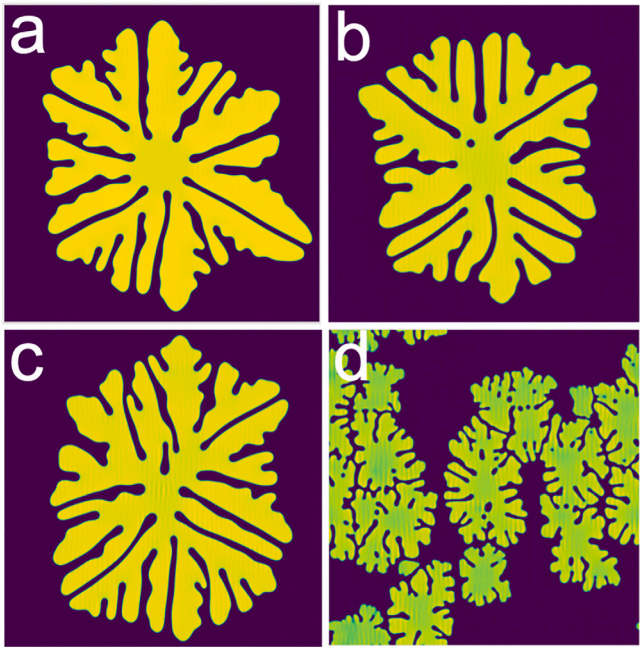


Fig. 8. Dendritic morphology at different noise intensities for $a_2 = -0.3$, $\bar{\psi} = -0.33$. Panel (a-d) correspond to noise intensities of 0.10, 0.30, 0.40 and 0.50, respectively, with $\lambda = 0.02(2\pi/a)$ and $k_{peak} = 0.1$.

At low noise intensity ($D = 0.10$), as shown in Fig. 8(a), the effect of noise on interface perturbation is minimal. The dendritic morphology remains largely symmetric, with fewer and more evenly distributed side branches, resulting in a more regular overall structure.

With moderate noise intensity ($D = 0.30$, 0.40), interfacial instability is significantly amplified, leading to increased branch density and a

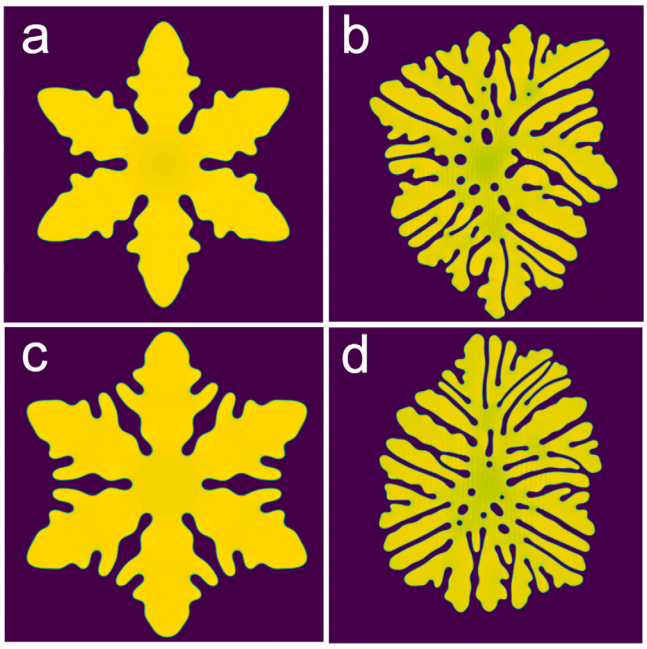


Fig. 9. (a, b) Dendritic morphologies at noise intensities of 0 and 0.20 for $a_2 = -0.3$, $\bar{\psi} = -0.329$. (c,d) Dendritic morphologies at noise intensities of 0 and 0.30 for $a_2 = -0.302$, $\bar{\psi} = -0.330$.

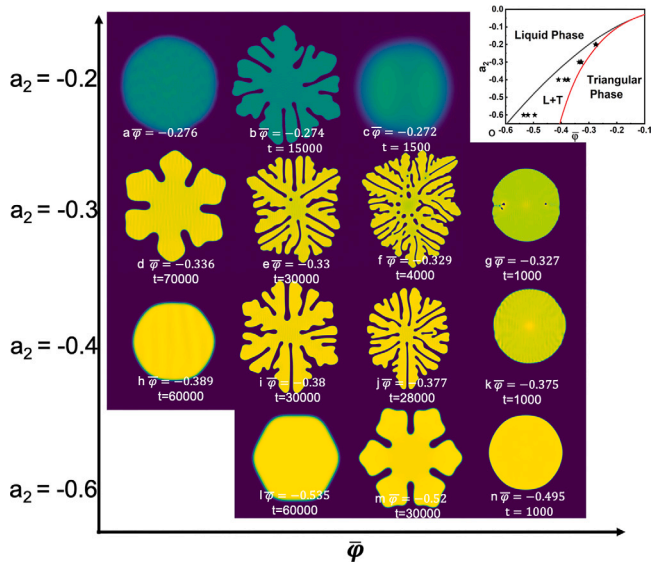


Fig. 10. Growth morphologies of crystals in different solid–liquid coexistence states under y-axis-biased colored noise at a noise intensity of 0.30, filter width $\lambda = 0.02(2\pi/a)$, and $k_{peak} = 0.1$.

more complex, asymmetric dendritic morphology (Fig. 8(b, c)). At this stage, noise promotes the formation of secondary branches through a positive feedback mechanism.

At high noise intensity ($D = 0.50$), the substantially increased noise energy strongly amplifies the amplitude of perturbations near the interface, resulting in a pronounced increase in the number of dendritic branches. Moreover, additional dendrite nucleation is observed in the liquid-phase region surrounding the original grain (Fig. 8(d)).

It should be emphasized that, although excessively strong noise amplifies perturbations at specific wavenumbers, the intrinsic linear stability properties of the interface remain unchanged. Consequently, this situation differs fundamentally from the case with $k_{peak} = 0$ shown in Fig. 6, where the instability is associated with an intrinsic alteration of the linear stability structure.

In summary, increasing noise intensity enhances interfacial instability, promoting more complex and asymmetric dendritic structures. Noise amplifies the randomness of interfacial perturbations, facilitating side-branch formation and greater morphological diversity. Further increasing the noise intensity may lead to interface instability, with dendrites also growing in the liquid phase.

Undercooling dependence of noise-regulated growth mechanisms. The regulatory effect of noise on dendritic growth is strongly dependent on the initial undercooling level. As shown in Fig. 9, the noise regulation mechanism differs significantly under varying contributions of interface energy anisotropy and interface kinetics:

When the interface kinetics anisotropy is high ($a_2 = -0.30$, $\bar{\psi} = -0.329$), the interface migration velocity becomes highly sensitive to local perturbations. Noise intensifies this sensitivity by introducing rapid and irregular fluctuations, which in turn amplify the effects of kinetic anisotropy. As shown in Fig. 9(a, b), increasing the noise intensity gives rise to numerous transient perturbations along the interface, which are rapidly amplified during the subsequent evolution. Both primary and secondary dendrite arms undergo frequent bifurcations, leading to a pronounced increase in branching density. Consequently, the resulting dendritic morphology becomes markedly more complex and asymmetric.

Increasing the interface energy anisotropy ($a_2 = -0.302$, $\bar{\psi} = -0.330$) enhances the sensitivity of the interface to perturbations. In the presence of noise, this heightened sensitivity amplifies the effect of random fluctuations, as the energy barrier between preferred and non-preferred growth directions becomes more pronounced. As shown in Fig. 9(c, d), at high noise intensity ($D = 0.3$), the coupling between strong interfacial energy anisotropy and stochastic perturbations markedly enhances the instability of dendritic growth. Consequently, the dendrites develop highly branched structures characterized by complex, multi-scale and multi-orientation morphologies.

These results indicate that the influence of noise on dendrite evolution arises from its complex interaction with interface energy and kinetic anisotropy. Increasing either type of anisotropy enhances the interface's susceptibility to perturbations, amplifying the effect of noise and driving morphological transitions from stable dendrites to complex asymmetric structures. Specifically, the stronger anisotropy broadens range of wavenumbers involved in the evolution process by simultaneously increasing both the peak value and the width of the dispersion relation $\alpha(k)$. These findings advance our understanding of noise-driven solidification and provide a theoretical foundation for controlling crystal growth microstructures in materials processing.

Finally, we analyze the evolution of dendrite morphology under varying combinations of interface energy anisotropy and interface kinetic anisotropy in the presence of noise, as shown in Fig. 10. The results reveal that in the steady-state growth regime, where interface energy dominates, noise perturbations are largely damped and induce minimal morphological change. By contrast, in the unsteady-state growth regime, dominated by interface kinetic anisotropy, noise amplifies growth instabilities by perturbing the solid–liquid interface, leading to pronounced morphological transformations. Moreover, noise significantly accelerates dendrite growth and increases interface roughness, as reflected by the emergence of complex side-branch bifurcations and surface microstructures. Noise applied along the y-axis induces a significant bias in dendrite growth, whereas the effect of noise along the x-axis is shown in Fig. 11. Notably, the morphology observed in Fig. 10(j) closely resembles tip splitting in the directional solidification of Al-Cu alloys [38], further highlighting the critical role of noise in regulating dendritic growth patterns.

In summary, numerical analysis of circular nucleation interfaces reveals that Gaussian colored noise plays a crucial role in regulating

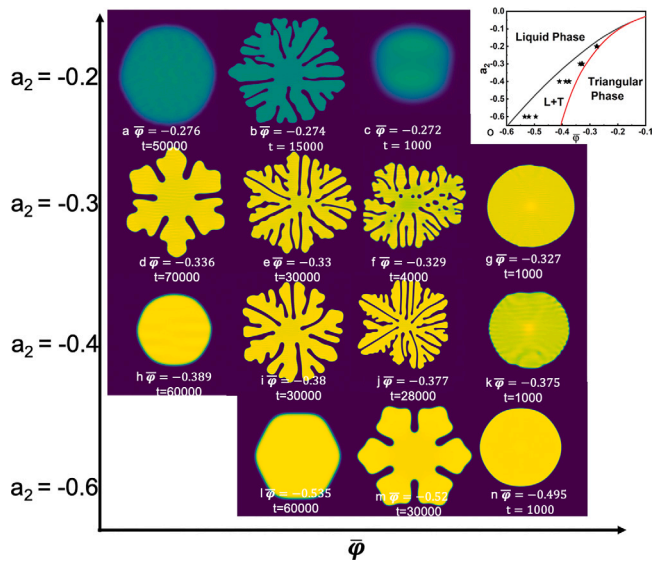


Fig. 11. Growth morphologies of crystals in different solid–liquid coexistence states under x-axis-biased colored noise at a noise intensity of 0.30, filter width $\lambda = 0.02(2\pi/a)$, and $k_{peak} = 0.1$.

dendrite morphology evolution. First, the noise's characteristic wave number selectively excites linear modes across different wave number ranges, controlling the spatial scale of interface perturbations and their amplification. This, in turn, determines the intensity of the interface's morphological response. Directionally biased noise further amplifies perturbations along specific orientations, inducing pronounced directional dependence in morphology and growth behavior. Secondly, the noise's filtering width governs the frequency distribution of perturbations, influencing dendrite trunk thickness, branch density, and surface roughness. Excessively wide or narrow bandwidths result in, respectively, overly disordered perturbations or damping-suppressed growth. Noise intensity directly affects the amplification and randomness of interface perturbations: low-intensity noise has limited morphological impact, moderate-intensity noise promotes secondary dendrite formation, while high-intensity noise markedly increases growth instability, producing more complex structures with denser branching. Finally, the regulatory effect of noise critically depends on the system state. When interface kinetic anisotropy dominates, the interface becomes more sensitive to perturbations, amplifying noise effects. This accelerates growth rates, increases surface roughness, and alters the symmetry of dendritic structures. Overall, noise interacts with solid–liquid interface instability, interface energy, and kinetic anisotropy to achieve diverse control over crystal morphology — from regular dendrites to complex asymmetries and even isotropic grains. This provides a clear theoretical basis for understanding how random perturbations influence crystal morphology during actual solidification processes.

4. Conclusions

In this study, the APFC model was employed to systematically investigate the kinetic behavior of crystal growth under different undercooling conditions, providing an in-depth analysis of dendritic growth kinetics and morphology control mechanisms. By incorporating Gaussian colored noise, this research further elucidates its critical role in regulating dendritic morphology and growth dynamics.

The results demonstrate that crystal growth under varying undercooling conditions leads to distinct dendritic morphologies, including interface-energy-driven dendrites, interface-kinetics-dominated dendrites, and tip-split dendrites. When interface energy anisotropy is weak, the morphology evolves from equiaxed crystals to hexagonal dendrites and ultimately to isotropic round crystals as interface kinetics become more dominant. In contrast, when interface energy anisotropy dominates, increasing contribution from interface kinetics drive sequential transitions from equiaxed crystals to hexagonal dendrites primarily controlled by interface energy, and eventually to isotropic round crystals. Under strong competition between interface energy anisotropy and kinetic anisotropy, complex tip-splitting dendrites may also emerge. These results highlight the versatility and advantages of the APFC model in modeling crystal growth, particularly its capability to capture mesoscale solid–liquid interface kinetics and microstructural evolution.

More importantly, Gaussian colored noise is incorporated into the APFC framework to systematically elucidate its role in regulating dendritic morphology, directional selectivity, and growth kinetics. Among the noise parameters, the characteristic wavenumber plays a central role by directly selecting the dominant length scale of interfacial perturbations. Through this mechanism, specific instability modes are preferentially amplified, leading to controlled symmetry breaking and directional growth during dendritic evolution. The noise intensity governs the amplitude of these perturbations, analogous to thermal or compositional fluctuations in experimental systems, while the filter width determines the spectral bandwidth over which unstable modes can be activated and interact with the crystal lattice. Together, these parameters regulate the extent to which intrinsic interfacial instabilities are expressed. However, it is the characteristic wavenumber that primarily dictates the spatial scale and orientation of emerging dendritic patterns, thereby acting as the dominant control parameter for morphology selection. Through this framework, the APFC model captures the selective coupling between noise spectra and intrinsic instability modes, providing a physically transparent description of how specific perturbation scales are amplified during solidification. Compared with conventional phase-field models, this approach highlights the pivotal role of spectral selectivity in noise-driven pattern formation and offers a powerful route for controlling microstructural evolution through targeted perturbations.

However, the APFC model employed in this study is restricted to pure single-component systems under isothermal conditions and does not incorporate effects such as temperature gradients, heat flux, or alloy composition. Future work may extend the present framework to non-isothermal and multi-component systems, enabling simulations that more closely reflect experimental conditions and allowing further exploration of dendritic evolution under the combined influence of noise, anisotropy, and external fields.

CRediT authorship contribution statement

Chen Li: Writing – original draft, Investigation, Data curation, Conceptualization. **Wenjun Zhu:** Writing – review & editing, Supervision, Resources. **Kun Wang:** Writing – review & editing, Methodology, Funding acquisition. **Xiaoping Ouyang:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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